Degala Chenchupradeep

(+91) 9347539941 | Pradeepdegala143@gmail.com | LinkedIn | GitHub: pradeepdegalaroyal | Portfolio

PROFESSIONAL SUMMARY

Ph.D. Researcher in Quantum Machine Learning with expertise in AI, Deep Learning, and Quantum Computing. Proficient in Python, Qiskit, and PennyLane. Develops quantum-inspired and gate-based models for many-body physics, error mitigation, and QAOA optimization, with applied work in quantum communication (QKD), underwater acoustic networks, and quantum-like fuzzy decision making. Ph.D. Scholar at SRM University AP. Publishes reproducible, simulation-based research projects as open-source repositories with pinned dependencies, tests and CI.

RESEARCH INTERESTS

- Quantum machine learning (QML) and variational quantum algorithms: QAOA and VQE.
- Quantum error mitigation and noise-aware algorithm design.
- Tensor networks (matrix product states) for many-body physics.
- Quantum communication and quantum key distribution (QKD).
- Underwater communication.
- Quantum photonics.
- Quantum information theory.

CORE SKILLS

- **Quantum Computing:** Qiskit, PennyLane, Cirq, Tensor Networks (MPS), ZNE (Error Mitigation), VQA, QAOA, VQE.
- **Machine Learning:** PyTorch, TensorFlow, Scikit-learn, Reinforcement Learning, NLP (BERT), GCN.
- **Tools & Languages:** Python, C++, SQL, Django, Flask, Pandas, NetworkX, Big Data (Spark, Hadoop), Cloud Computing.

RESEARCH PROJECTS

Recent open-source research projects (newest first)

- **Dephasing tolerance of quantum-annealing exponents in spin glasses** *GitHub: quantum-annealing-exponent-dephasing-spin-glass | Nov 2026*
  Stack: Python, PyTorch, NumPy, pandas, NetworkX, matplotlib, pytest, GitHub Actions.
  – Built a GPU state-vector annealing simulator with a Z-dephasing trajectory model checked against an exact density matrix (0.2009 ± 0.0011 vs 0.1993 residual energy).
  – Showed the coherent residual-energy exponent exceeds tuned simulated annealing in 15 of 15 seed ensembles at n = 8, 10, 12 (e.g. 1.58 ± 0.08 vs 0.51 ± 0.07 at n = 12), while it falls with n.
  – Measured the break-even dephasing rate gamma* = 0.00092, 0.00157, 0.00169 for n = 6, 8, 10 (refuting a decrease with n); dephasing cut the n = 10 exponent from 2.02 to 0.84, a 10 % coupling error did not.
- **Measurement-free vs feed-forward QEC on a trapped-ion memory** *GitHub: measurement-free-vs-feedforward-ion-memory | Nov 2026*
  Stack: Python, NumPy, pandas, Matplotlib, Qiskit, Qiskit Aer, pytest, GitHub Actions.
  – Built an exact probability-tensor simulator of a three-ion repetition memory with feed-forward and measurement-free correction; Qiskit Aer density matrices agree with it to 3.1e-16 for an explicit Toffoli decomposition.
  – Mapped the break-even measurement time over two-qubit error, idle, readout and spectator noise: 18.4 to 84.0 two-qubit gate-times for a six-gate Toffoli, 2.6 to 24.2 for a native one; measurement-free needs 9 ions vs 5 at 3 cycles.
  – Quantified experimental cost and tolerance: 44,298 shots per scheme to resolve the winner at tau_m = 32.8, and 7.19 % +/- 2.84 % wrong scheme choices under a 0.3 log-normal noise-model error (5 seeds).
- **Erasure-check scheduling for a dual-rail repetition-code memory** *GitHub: erasure-check-period-dual-rail-repetition-memory | Nov 2026*
  Stack: Python, NumPy, pandas, Matplotlib, pytest, GitHub Actions.
  – Built an exact Viterbi minimum-weight space-time decoder (equivalent to matching) for a dual-rail repetition code with flag windows of unknown erasure time; graded priors beat calibrated flat priors by 22.2 ± 0.7 % at period 5.
  – Mapped where checking less often than every round wins on a 5x5 check-duration/back-action grid: best period above 1 in 16 of 25 cells, above 2 SE in 10 cells, maximum gain 25.7 %.
  – Showed the logical error rate per microsecond falling from 5.282e-3 (d=3) to 0.892e-3 (d=5) to 0.134e-3 (d=7) with checks every round, against 18.272e-3 for classical majority vote at d=5.

- **Low-rank digital twin for sentinel-based qubit recalibration** *GitHub: low-rank-drift-twin-sentinel-recalibration* | *Nov 2026*
  Stack: Python, NumPy, SciPy, pandas, Matplotlib, Qiskit, Qiskit Aer, pytest, GitHub Actions.
  – Built a simulated many-qubit drift model and a rank-selected SVD/Kalman digital twin driven by 16 sentinel qubits; at N=4096 it needed 380.5 ± 6.6 shots/qubit/hour vs 915.0 ± 33.3 for periodic Ramsey recalibration to hold mean infidelity 3e-4 (5 seeds).
  – Mapped the scaling and failure regime: reduction 0.27 ± 0.01 at 16 qubits, 1.62 ± 0.03 at 256 and 2.51 ± 0.06 at 16,384 qubits; no saving when the correlated drift share is 0.5 or lower.
  – Validated building blocks against Qiskit (gate infidelity, max abs diff 2.6e-16) and Aer-sampled Ramsey circuits (std ratio 0.92-1.02); ablations over sentinel count, loadings and placement, with unit tests and CI.
- **Learned code design for the quantum switch of depolarising channels** *GitHub: cyclic-order-switch-depolarizing-control-design* | *Nov 2026*
  Stack: Python, PyTorch, NumPy, pandas, Matplotlib, pytest, GitHub Actions.
  – Built an exact Kraus-tensor simulator of a quantum switch of 2-5 completely depolarising qubit channels (cyclic and all-permutation orders) and a PyTorch encoder-decoder optimiser; learned code gives Fe 0.6250 against 0.3750-0.4500 for the fixed protocol and the 0.5 classical limit.
  – Showed by ablation that decoder-only learning stays at the fixed-protocol value (0.4356 vs 0.4375 at N=4) while using all 24 orders of four channels lifts the best restart to Fe 0.7100 (mean 0.6526 over 3 restarts).
  – Swept control dephasing and control replacement and found the margin falls as 0.625 - 0.125 lambda and 0.625 - 0.375 lambda with no gain from re-design; certifying fidelity above 2/3 needs 216 shots at zero noise and 21,569 at 0.3 replacement.
- **Shot Cost of Non-Markovian Memory Witnesses on a Qubit** *GitHub: shot-cost-of-non-markovian-memory-witnesses* | *Nov 2026*
  Stack: Python, NumPy, pandas, scikit-learn, Qiskit, Qiskit Aer, Matplotlib, pytest, GitHub Actions.
  – Turned the Breuer-Laine-Piilo revival criterion into a bootstrap max-rise hypothesis test on finite-shot data: false-alarm rate at most 0.047 on Markovian baths versus at least 0.969 for the raw any-increase rule (4 seeds x 80 runs).
  – Fitted shots-to-80%-power against exact revival depth for four witnesses: log-log slopes -1.90 (full tomography), -1.74 (matched X,Y) and -1.59 (relative entropy); relative entropy costs 3.7 to 6.8 times more shots than trace distance.
  – Swept hardware-style noise: readout flip 0.1 raises matched-design shots 1.63x (predicted 1.56x); a single-X witness false-alarms at rate 1.0 under frame detuning while the X,Y design stays at 0.0; Qiskit Aer check within 0.0058.
- **Learned Trotter circuits costed in fault-tolerant T gates** *GitHub: learned-trotter-circuits-t-count-synthesis-budget* | *Nov 2026*
  Stack: Python, PyTorch, NumPy, SciPy, pandas, Matplotlib, pytest, GitHub Actions.
  – Learned all 91 rotation angles of a Strang-shaped circuit for a 6-qubit disordered Ising propagator (t = 1): spectral-norm error 0.0247 ± 0.0057 versus 0.1005 ± 0.0131 for Strang at equal rotation count (5 instances).
  – Costed circuits in Clifford+T gates with an additive synthesis-error model: at error target 0.1 learned circuits need 2238 ± 166 T gates versus 3795 ± 208 for the best Strang/Suzuki formula; ratio 1.16-2.19 over n = 4-8, t = 0.5-1.5.
  – Ablation and noise sweeps: random start 0.1236 ± 0.0410 versus 0.0247 for Strang start; three shared scales per step narrowly miss the 2x hypothesis (1.96); Pauli p = 0.001 infidelity 0.0651 versus 0.1901 for equal-error Strang.
- **Blinking-aware g2(0) certification for single-photon sources** *GitHub: blinking-aware-g2-certification-photon-source* | *Nov 2026*
  Stack: Python, NumPy, SciPy, pandas, Matplotlib, pytest, GitHub Actions.
  – Built a validated Hanbury Brown-Twiss simulator (peak-area model checked against a pulse-level simulation) and an envelope-fitting g2(0) estimator; at b = 1, tau_b = 30 RMSE is 0.0119 vs 0.0205 for the outer-peak baseline, coverage 0.957 vs 0.680 for the all-peaks baseline.
  – Measured counts to certify g2(0) < 0.05 for a 0.03 source under moderate blinking: 6400 per side peak for the envelope estimator in 5/5 seeds, versus 12800 (4/5 seeds) for the all-peaks baseline and never for the outer-peak baseline.
  – Quantified certificate fragility: a -0.02 error in the background fraction gives a false-certification rate of 0.872 (envelope) vs 0.463 (outer-peak baseline) for a g2(0) = 0.06 source; the benefit also fails when blinking is slower than the record (coverage 0.532).

- **Shot-cost break-even for Rydberg-atom MIS sampling and tier routing** *GitHub:* *rydberg-mis-shot-cost-tier-routing* | *Nov 2026*
  Stack: Python, NumPy, SciPy, pandas, scikit-learn, Matplotlib, pytest, GitHub Actions.
  – Built an exact state-vector simulator of an adiabatic Rydberg sweep (1/r^6 interaction, readout noise) and measured shots-to-99% success for maximum independent set on 120 unit-disk graphs, 8-14 atoms, 5 seeds; analog sweep log10 S99 0.389 ± 0.054 (n <= 12).
  – Benchmarked against penalty QAOA (p=1: 0.663 ± 0.039; p=2: 0.681 ± 0.084) and three classical samplers (min-degree greedy 0.314 ± 0.036); the sweep needed strictly fewer shots than the best classical sampler on 17.5 % of instances.
  – Defined a break-even shot cost (median 10.0 classical site updates) and showed by ablation that greedy post-processing lifts per-shot success under 1/r^6 from 0.409 to 0.819; a 64-sample pre-screen router cost 30.0x more than always running classically.
- **Local vs Global Cost under Noise: Barren-Plateau Shot Cost** *GitHub:* *local-vs-global-cost-noisy-barren-plateau-shots* | *Nov 2026*
  Stack: Python, PyTorch, NumPy, pandas, matplotlib, pytest, GitHub Actions.
  – Built a batched exact density-matrix simulator (RY + CZ ansatz, depolarising noise, n = 3-8) and measured gradient-variance scaling: global-cost slope -1.77 +/- 0.07 versus local -0.65 +/- 0.01 at logarithmic depth (3 seeds).
  – Quantified the shot cost of resolving one gradient: at n = 8 the global cost needed 123 +/- 53 shots versus 16.7 +/- 1.1 for the local cost at p = 0.02, and noise steepened the global linear-depth slope by -0.39 versus -0.18 for local.
  – Ran equal-shot-budget training against SPSA and random-search baselines with pre-stated hypotheses; reported two falsified (local slope -0.65; Adam local 0.070 vs global 0.038 median infidelity at n = 8) and a post-hoc 1/n^2 prefactor check.
- **Adaptive vs hardware-efficient VQE under noise-induced barren plateaus** *GitHub:* *adapt-vs-hea-noise-induced-barren-plateau-vqe* | *Nov 2026*
  Stack: Python, PyTorch, NumPy, SciPy, pandas, Matplotlib, pytest, GitHub Actions.
  – Built a real-valued PyTorch density-matrix simulator with CNOT-attached depolarising noise; fitted RMS-gradient decay for depth-n hardware-efficient VQE: -0.150 per qubit at p=0 versus -0.2615 at p=0.01 (depth-2: -0.0049 vs -0.0063).
  – Implemented noisy-device ADAPT-VQE with exact parameter-shift operator selection; a measured-energy stopping rule cut error from 0.1224 to 0.0401 at n=7, p=0.01, and beat best-depth HEA only at n=7 (0.0296 vs 0.0304 at p=0.005).
  – Benchmarked against a zero-CNOT mean-field baseline over 3 seeds: at p=0.01 the best circuit improves error only from 0.0407 to 0.0385 at n=7; pre-registered hypothesis H2 was falsified and reported as such.
- **Power-mean error budgets for a repetition-code memory** *GitHub:* *power-mean-error-budget-repetition-code-memory* | *Nov 2026*
  Stack: Python, NumPy, SciPy, pandas, NetworkX, Matplotlib, pytest, GitHub Actions.
  – Built a detector-error-model simulator and exact-matching decoder for a circuit-level repetition-code memory (distance 3 and 5) with gate, readout and idle noise, in NumPy with no stabiliser package; 10 unit tests and continuous integration.
  – Showed the standard additive error budget gives 0.16 to 0.36 of the simulated logical error, while a power-mean budget (sum of eps^(1/t))^t with t=(d+1)/2 gives 0.86 to 0.99 (5 seeds, 7 operating points).
  – Mapped validity over 36 noise-scale and readout-ratio cells: one calibration predicts 30 of 36 within 25 percent (additive: 0 of 36); rejected the cost hypothesis, since direct Monte Carlo had lower error at every shot budget.
- **Noise bias vs variance in federated quantum classifiers** *GitHub:* *noise-bias-vs-variance-federated-vqc-aggregation* | *Nov 2026*
  Stack: Python, PyTorch, NumPy, SciPy, scikit-learn, Matplotlib, Qiskit (validation), pytest, GitHub Actions.
  – Built an exact density-matrix simulator (noise as 256x256 superoperators, checked against Qiskit) and a federated parameter-shift training loop for a 4-qubit, 16-parameter classifier on heterogeneous noisy clients with binomial shot noise.
  – Separated device noise into bias (calibrated attenuation correction, 16.2% of the shot budget) and variance (Fisher weights): correction alone gave test MSE 0.440 ± 0.015 vs 0.465 ± 0.043 for FedAvg at 8 shots, 4 of 5 seeds, paired p = 0.12.
  – Falsified the inverse-variance weighting hypothesis: MSE rose with weight exponent (0.440 at alpha 0 to 0.506 at alpha 4 for corrected clients); ran ablations, noise, calibration-error and label-skew sweeps over 195 runs, 5 seeds, with CI.

- **Noise-aware MPDO simulation of noisy Trotter quench circuits** *GitHub:*
  noise-aware-mpdo-bond-dimension-trotter-quench | Nov 2026
  Stack: Python, NumPy, SciPy, pandas, Matplotlib, Qiskit, Qiskit Aer, pytest, GitHub Actions.
  – Built a noise-aware matrix-product density-operator (TEBD) simulator for Trotterised Ising quench circuits and validated it against Qiskit Aer density-matrix simulation to a maximum deviation of 1.6e-15 (results/validation.json).
  – Measured fixed-tolerance bond dimensions over 6-12 qubits and 3 seeds: the noisy-circuit MPDO is cheaper than the ideal-circuit MPS only above two-qubit error p2 ~0.084 (n=8), and about 2.9e3 times costlier at p2=0.01.
  – Showed a noise-blind ideal-circuit check deviates from the noisy output by 0.097 +- 0.009 at p2=0.01 (n=8), ten times the 10^4-shot noise; two-qubit depolarising noise removed 1.48 of 1.70 decades of truncation error in the ablation.
- **Erasure-invisible loss bias in dual-rail memory: echo vs inversion** *GitHub:*
  dual-rail-erasure-bias-swap-echo-vs-inversion | Oct 2026
  Stack: Python, NumPy, SciPy, pandas, Matplotlib, pytest, GitHub Actions.
  – Derived and verified a (2 eta T)^2/24 scaling for the erasure-invisible conditional infidelity of an asymmetric-loss dual-rail memory; fitted log-log slope 1.999 at hold 0.1, maximum deviation 4.1 percent.
  – Compared bare postselection, calibrated no-jump inversion and a Floquet-style swap echo with shot noise over 5 seeds: Bloch-vector MSE at hold 0.3 was 1.56e-2 (bare), 7.63e-4 (inversion), 1.21e-3 (echo).
  – Mapped the echo-versus-inversion crossover over calibration drift and pulse over-rotation: echo wins in 51 of 81 grid cells, only for over-rotation below 0.018, and needs drift of at least 0.125 at over-rotation 0.0039.
- **Shot-budgeted Nystrom approximation of quantum kernels** *GitHub:*
  shot-budget-nystrom-quantum-kernel-noise-floor | Oct 2026
  Stack: Python, Qiskit, Qiskit Aer, NumPy, SciPy, scikit-learn, pandas, Matplotlib, pytest, GitHub Actions.
  – Built a shot-accounted quantum-kernel simulator (Qiskit-verified to 4.2e-15, Aer noise check): at 3.2e6 training shots Nystrom with CV-chosen landmarks scored 0.898 ± 0.064 vs 0.829 ± 0.102 for the full-kernel SVM (6 qubits, 2 datasets, 5 seeds).
  – Over 36 qubit/gate-error/budget cells, plain Nystrom was significantly better than the full-kernel SVM in 9 cells and worse in 0 (mean +3.366 points); the exact quantum kernel (0.959 ± 0.027) stayed below a tuned RBF SVM (0.969 ± 0.019).
  – Pre-registered a spectral noise-floor cut and reported its failure: cut factor 1 gave -0.733 ± 1.055 points at m = 40, 80 (rejected), while post-hoc factor 0.25 gave +1.204 ± 0.344; landmark-count optimum moved from m = 5 to m = 20 with budget.
- **QCNN vs string order: shot-budget benchmark for phase recognition** *GitHub:*
  qcnn-vs-string-order-spt-shot-budget | Oct 2026
  Stack: Python, PyTorch, NumPy, SciPy, scikit-learn, Qiskit, pandas, Matplotlib, pytest, GitHub Actions.
  – Built a simulated benchmark of a trained quantum convolutional neural network against a string-order threshold and a four-feature logistic regression for topological-phase recognition; at 10 state copies, 16 qubits, accuracy near criticality is 0.802 ± 0.019 vs 0.727 ± 0.000.
  – Showed the advantage reverses with more copies: at 8 qubits and 3000 copies the string estimator reaches 0.965 ± 0.023 vs 0.879 ± 0.057 for the circuit (5 seeds), falsifying my pre-stated shot-efficiency hypothesis.
  – Swept gate noise against readout noise with exact density matrices (+11.7 ± 4.8 points for the circuit at 5% readout flip, -3.2 ± 3.7 at 5% gate error) and probed tensor-network cost: bond dimension 4 gives accuracy 0.979, equal to the exact input.
- **Entanglement-entropy features vs distance baselines (QML study)** *GitHub:*
  entanglement-features-vs-distance-baselines | Oct 2026
  Stack: Python, NumPy, SciPy, scikit-learn, pandas, Matplotlib, pytest, GitHub Actions.
  – Built a batched six-qubit statevector simulator and random-Pauli snapshot estimator for entanglement-entropy and mutual-information features; tested against dense operators in 8 pytest tests, with continuous integration.
  – Compared the descriptor with three classical baselines and a fidelity quantum kernel under identical cross-validation over 5 seeds: it beat the best classical baseline in 2 of 32 signal cells, by at most 1.07 percentage points.
  – Priced the descriptor in circuit executions: the median mutual-information error is 0.023 nats at 6400 snapshots, and a model trained on noiseless features falls from 0.632 to 0.535 accuracy at 0.2 percent two-qubit error.

- **Gate budget of a constrained-mixer QAOA for portfolio selection** *GitHub: hypercube-mixer-qaoa-knapsack-gate-budget | Oct 2026*
  Stack: Python, Qiskit, Qiskit Aer, NumPy, SciPy, pandas, Matplotlib, pytest, GitHub Actions.
  – Built Boolean-minimised (Shannon cube cover) compilation of a feasibility-preserving QAOA mixer, verified against Qiskit circuits; CX per layer fell from 31,384 to 6,831 at 8 items and from 246,042 to 47,979 at 10 items.
  – Benchmarked against slack-penalty QAOA (137 CX per layer at 8 items) and classical controls over 5 seeds; the constrained mixer won only at 10 items (regret 4.86% vs 11.29%, noise-free, p = 2) and lost to annealing (3.99%).
  – Validated a gate-count depolarising model against Aer density-matrix runs (feasible fraction 0.796 vs 0.735 at error 1e-2) and reported break-even errors at 10 items with median 1e-5 (vs uniform feasible sampling) and 6.5e-5 (vs penalty QAOA, p = 1).
- **Kernel-aware DFS pairing under correlated dephasing** *GitHub: kernel-aware-dfs-pairing-correlated-dephasing | Oct 2026*
  Stack: Python, NumPy, SciPy, pandas, Matplotlib, Qiskit, pytest, GitHub Actions.
  – Derived a closed-form logical idle fidelity for correlated Gaussian dephasing and enumerated all 105 to 10395 qubit pairings; validated against sampled phases and a Qiskit CNOT encoder (0.8086 exact vs 0.8047 Qiskit).
  – Showed kernel shape is irrelevant to the compile decision (all 16 true/assumed kernel pairs retain at least 0.995 of oracle gain) while a 1-pitch geometry error leaves geometry-only compilers at 0.350 ± 0.097.
  – Measured calibration cost: 300 Ramsey shots per pair retain 0.858 ± 0.108 of oracle gain at 1-pitch mismatch, falling to 0.804 ± 0.102 at 10 qubits; mapped break-even CNOT error against a CPMG baseline.
- **Physics-fitted vs GAN sim-to-real for charge stability diagrams** *GitHub: physics-fitted-vs-gan-charge-stability-sim2real | Oct 2026*
  Stack: Python, PyTorch, NumPy, SciPy, pandas, Matplotlib, pytest, GitHub Actions.
  – Built a constant-interaction double-dot simulator with a measurement layer (thermal broadening, sensor peak, 1/f drift, charge jumps) and exact transition-line labels; fitted 11 parameters to unlabeled images by differential evolution on 24 image statistics.
  – Compared the fit with a CycleGAN-style translator and hand augmentation over 8-128 unlabeled images and 3 artefact strengths: fitting closed 0.78 ± 0.13 of the source-to-oracle Dice gap at strength 0, but -0.08 ± 0.33 at strength 0.5.
  – Showed realism and utility diverge: a longer GAN run lowered RF-FD from 0.14 ± 0.02 to 0.09 ± 0.02 while Dice fell from 0.440 ± 0.007 to 0.395 ± 0.008; ablation ranked charge jumps as the most important component (Dice 0.314 ± 0.043 without).
- **Anytime-valid shot stopping for noisy quantum circuit tests** *GitHub: anytime-valid-shot-stopping-quantum-circuit-tests | Oct 2026*
  Stack: Python, Qiskit, Qiskit Aer, NumPy, SciPy, pandas, Matplotlib, pytest, GitHub Actions.
  – Built a sequential pass/fail test for quantum circuits with a finite-sample Hoeffding bound; on 900 exact-noise 4-qubit cases it committed 0.000 wrong verdicts within 25% of the tolerance versus 0.136 +- 0.044 for a stability stopping rule.
  – Measured the price of the guarantee: abstain share 0.815 +- 0.053 near the tolerance, and mean shots on correct circuits rising from 1789 to 8719 between 3 and 6 qubits (stability stop: 1904 to 3588).
  – Showed that a chi-square assertion declares correct noisy circuits buggy on 0.916 +- 0.013 of clear-correct cases, and that an asymptotic confidence radius with repeated peeking breaks the 0.05 level (0.118 +- 0.017).
- **Retry-window budget for photonic fusion networks** *GitHub: fusion-retry-window-delay-loss-budget | Oct 2026*
  Stack: Python, NumPy, SciPy, numba, pandas, Matplotlib, NetworkX, pytest, GitHub Actions.
  – Built a union-find winding-cycle kernel (numba) for erasure correctability of a periodic cubic fusion network, L = 4 to 10, 5 seeds; measured crossing 0.2493 ± 0.0023 and checked it against an independent breadth-first lift in tests.
  – Modelled repeat-until-success fusion with a synchronous-clock delay-loss term and found a finite optimal window: 14 slots at 0.1 dB per slot (2.343 ± 0.007 dB loss budget, success 0.75) versus 1.671 dB at 30 slots.
  – Showed single-attempt fusion (success 0.5 and 0.75) cannot reach 5 percent logical failure, and that whole-node outages cut the budget from 2.343 dB to 1.775 dB; a lattice-blind rule matched the optimum in 35 of 35 cells.
- **Software recovery of loss in photonic Mach-Zehnder meshes** *GitHub: lossy-mzi-mesh-relabelling-vs-reprogramming | Oct 2026*
  Stack: Python, PyTorch, NumPy, SciPy, pandas, Matplotlib, pytest, GitHub Actions.
  – Built exact Clements and Reck decompositions with a phase-dependent lossy-MZI model (error below 1e-10 lossless) and a differentiable PyTorch forward model for loss-aware phase re-optimisation.
  – Measured on 8 modes at 1 dB mean arm loss over 5 seeds: relabelling ports raises loss-inclusive fidelity by +1.77 ± 0.67 %, reprogramming phases by +0.52 ± 0.60 %, both by +1.99 ± 0.71 %.
  – Showed the gain trades heralded fidelity (-1.22 ± 0.29 pp) for transmission, and that relabelling keeps +0.84 ± 0.21 % under a log-normal calibration error of 2 while re-optimisation falls to -23.94 ± 20.19 %.

- **Shot and Noise Cost of Certifying Variational Quantum Linear Solvers** *GitHub: vqls-certificate-shot-cost-noise-floor | Oct 2026*
  Stack: Python, NumPy, SciPy, pandas, Matplotlib, pytest, GitHub Actions.
  – Derived exact delta-method moments of the Variational Quantum Linear Solver cost estimator and checked them against binomial shot simulation (issue rates 0.945 to 0.9545 against a nominal 0.95).
  – Measured shots to certify a precision target growing with a log-log slope of 4.74 Â± 0.23 in condition number (target 0.1); variance-optimal shot allocation cut shots 2.17 Â± 0.26 times, rising from 1.71 to 2.75 times between 3 and 6 qubits.
  – Mapped the certificate noise floor: raw measurement tolerates a per-CNOT error of 3e-6 for target 0.2, zero-noise extrapolation 3e-4 (100 times) at a 15.7-fold shot cost; conjugate gradient needs about 2.4e3 operations for the same system.
- **Noise-induced bias in quantum-proposal MCMC for gauge theory** *GitHub: noisy-quantum-proposal-mcmc-gauge-bias | Oct 2026*
  Stack: Python, PyTorch, NumPy, SciPy, pandas, Matplotlib, pytest, GitHub Actions.
  – Built an exact density-matrix simulator (4-10 qubits) giving noisy transition matrices of a quantum-proposal Metropolis chain for a Z2 gauge-Higgs model; raw hardware-style noise biases the energy by 0.153 +/- 0.006 thermal standard deviations.
  – Showed which noise preserves detailed balance (unital Pauli noise placed symmetrically, symmetric readout on both sides); gate Pauli-twirl, readout flip-twirl and matched input flips cut the bias to 0.00405 +/- 0.00102, a 37.9-fold reduction.
  – Swept amplitude damping (0 to 1.6%), Higgs hopping and lattice size; the tailored chain keeps spectral gap 0.0392 +/- 0.0043, 4.7x the classical single-flip chain, and a 300,000-step sampled chain gave -0.1610 against the exact -0.1526.
- **Cat-qubit magic state factory costing under bit-flip floor** *GitHub: cat-qubit-magic-state-factory-costing | Oct 2026*
  Stack: Python, NumPy, SciPy, Matplotlib, pytest, GitHub Actions.
  – Enumerated the 15-to-1 distillation filter exactly over all 2^15 phase-flip patterns with noisy readout and validated it by sampling over 5 seeds (largest standardised deviation 1.76; leading coefficient 35.1).
  – Searched 8355 factory designs (cat size, 1 to 3 levels, distances up to 25): the cheapest design reaching 1e-15 output error uses cat size 20, distances 3-5-15 and 3.83e5 qubit-cycles per output state.
  – Showed a phase-flip-only design recipe misses its target in 188 of 196 phase-map cells at cat size 8; at target 1e-9 its true error is 1.42e5 times the target.
- **Compiler-aware noise budget of QAOA on a line device** *GitHub: commuting-qaoa-line-compiler-noise-budget | Oct 2026*
  Stack: Python, Qiskit, Qiskit Aer, NumPy, SciPy, pandas, Matplotlib, NetworkX, pytest, GitHub Actions.
  – Built a commutation-aware router for the QAOA cost layer on a qubit line (free term order, SWAP-interaction fusion, 16 restarts); mean cx count 132.8 -> 102.5 vs Qiskit level 3 best-of-16 over 15 MaxCut graphs, n=8, p=2.
  – Evaluated every compiled circuit exactly with density-matrix noise: normalised cut gain 0.409 -> 0.452 at 0.01 two-qubit error (paired Wilcoxon p = 0.0009); on dense graphs (density 0.8) the p=2 vs p=1 break-even error rose from 0.0066 to 0.0105.
  – Ran a four-step ablation, a swap-network baseline, a scaling check to n=10 (cx 216.6 -> 167.8) and a classical local-search reference (gain 0.855); reported the negative result that density-0.3 graphs show no reliable gain.
- **Quantum vs classical classifiers on label-scarce fraud-like data** *GitHub: label-scarce-fraud-qml-vs-classical | Oct 2026*
  Stack: Python, PyTorch, NumPy, scikit-learn, pandas, matplotlib, joblib, pytest, GitHub Actions.
  – Built a batched 4-qubit statevector simulator in PyTorch and benchmarked a quantum kernel SVM and a 50-parameter data re-uploading circuit against 4 tuned classical models over 5 seeds and 32-256 labels on 15%-positive data.
  – Pre-registered 3 hypotheses and reported them honestly: the quantum models did not beat the RBF-SVM on fraud-like data (AP 0.560 vs 0.686 at n = 64), while a quantum-feature-map control confirmed the pipeline can detect a benefit (+0.194 AP at n = 256).
  – Ran ablation and noise sweeps: removing the entangling ring dropped AP from 0.649 to 0.327, and a Jacobian effective-rank diagnostic gave 9.9 of 50 circuit parameters vs 14.6 of 49 for a matched MLP.

- **Loss-optimal switch trees for photon-source multiplexing** *GitHub: loss-optimal-switch-tree-photon-multiplexer | Oct 2026*
  Stack: Python, NumPy, pandas, Matplotlib, pytest, GitHub Actions.
  – Derived an exact O(N^3) dynamic program over all switch-tree shapes for thermal heralded sources; at 32 sources and 0.316 dB per switch the optimal shape gains 5.3 percent in one-photon probability over a balanced tree, and 60.1 percent at 128 sources and 1 dB.
  – Measured the best infidelity versus per-switch loss from 0.01 to 1 dB: log-log slopes 0.51 (optimal shape), 0.52 (balanced) and 0.38 (cascade); closed forms agree with Monte Carlo (0.5847 exact versus 0.5846 +/- 0.0010).
  – Benchmarked a generic hill-climbing shape search against the exact optimum over 5 seeds: gap 0.73 percent at 32 sources and 3.26 percent at 64; a design-mismatch map shows the optimal shape beats a matched balanced design in 73.5 percent of cases.
- **Shot-fair benchmark of damped quantum natural gradient VQE** *GitHub: shot-fair-damped-qng-vqe-benchmark | Oct 2026*
  Stack: Python, NumPy, SciPy, pandas, Matplotlib, pytest, GitHub Actions.
  – Built a shot-ledger VQE benchmark (NumPy statevector, 6-qubit Ising ring, 48 parameters) charging gradient, metric and trust-region circuits; at 32 shots per setting standard QNG reached 26.87 % median error, other methods at most 2.79 %.
  – Damped QNG with fixed damping 0.1 first reached 5 % energy error after 512,000 median shots versus 1,105,920 for parameter-shift Adam and 1,140,000 for SPSA (5 seeds); final errors 1.87 +- 0.07 % vs 2.03 +- 0.10 % for Adam.
  – Pre-stated two hypotheses with bootstrap tests: trust-region QNG failed the shot-fair advantage test (error ratio 0.83, interval 0.69 to 1.04) while the stability claim held on an exactly redundant ansatz (ratio 2.89, interval 1.55 to 4.21).
- **Imperfect erasure-flag decoding of the toric code** *GitHub: imperfect-erasure-flag-decoding-toric-code | Oct 2026*
  Stack: Python, NumPy, SciPy, NetworkX, PyTorch, Matplotlib, pytest.
  – Built a toric-code erasure simulator and exact minimum-weight perfect-matching decoder (L=3-7, 3 seeds); at flag efficiency 1.0 the logical failure probability is 0.0040 versus 0.0809 with flags ignored.
  – Showed that recalibrating the unflagged edge prior for hidden erasures is a no-op for matching: naive and calibrated decoders differ by at most 0.002 over the efficiency sweep, falsifying the hypothesis; the reason is uniform-weight scale invariance.
  – Benchmarked a flag-aware multilayer perceptron decoder against matching at L=5: it reaches 0.0724 at flag efficiency 1.0 versus 0.0040 for matching, a reported negative result.
- **QAOA portfolio selection under estimation error** *GitHub: portfolio-qaoa-estimation-error-floor | Oct 2026*
  Stack: Python, NumPy, SciPy, pandas, Matplotlib, pytest.
  – Simulated penalty and Dicke-state XY-mixer QAOA (12 qubits, 25 shots) for cardinality-constrained portfolio selection, scored out of sample against true market moments over 6 seeds.
  – XY-mixer QAOA (depth 2) at gate error 0.003 has extra out-of-sample regret 0.050 ± 0.056 at 24 months, refuting the 0.02 target; penalty QAOA gives 0.115 ± 0.093.
  – Exact-solver regret falls from 0.115 ± 0.114 (24 months) to 0.008 ± 0.014 (600 months), so optimisation error becomes the dominant loss as history grows.
- **Joint-fit interleaved randomized benchmarking study** *GitHub: joint-fit-interleaved-rb-coherent-error | Oct 2026*
  Stack: Python, NumPy, SciPy, pandas, Matplotlib, pytest, GitHub Actions.
  – Built an exact Pauli-transfer-matrix simulator of single-qubit interleaved randomized benchmarking with mixed coherent and T1/T2 noise at fixed infidelity.
  – Showed a joint fit with shared SPAM parameters and binomial weights beat standard IRB in 36 of 36 grid cells (mean RMSE ratio 0.536, range 0.394 to 0.708) over 5 seeds.
  – Ran ablation and readout sweeps: at 0.10 readout flip probability RMSE was 1.48e-3 for IRB versus 0.97e-3 for the joint fit; falsified my own high-shot hypothesis.
- **Noise-aware QAOA for satellite observation scheduling** *GitHub: satellite-scheduling-qaoa-cvar-noise | Oct 2026*
  Stack: Python, NumPy, SciPy, pandas, Matplotlib, pytest, GitHub Actions.
  – Built a statevector QAOA scheduler for max-weight independent set (n=8-12) with finite-shot COBYLA training and a CNOT-count-dependent depolarising noise model, versus random, annealing and greedy baselines.
  – Tested CVaR (alpha=0.1) against expectation training over 5 instances and 5 noise levels: repaired ratio 0.937 vs 0.929 at 0.003 CNOT error, within one std, so the hypothesis was rejected.
  – At n=12 QAOA put 0.360 probability mass on feasible schedules versus 0.025 for uniform sampling, and beat random-30 (0.923 vs 0.828 repaired ratio), but greedy reached 0.965.

- **Break-even margin map of a repetition-code quantum memory** *GitHub: repetition-code-breakeven-map | Oct 2026*
  Stack: Python, NumPy, pandas, Matplotlib, pytest, GitHub Actions.
  – Built a Monte Carlo of a bit-flip repetition-code memory with T1-twirled idling, gate and readout errors and an exact Viterbi space-time decoder; d=3 gave a logical/physical rate ratio of 0.134 at a 1 µs cycle.
  – Swept cycle time (0.5-4 µs) and ancilla readout error (0.005-0.1): worst-corner ratio was 0.391 (d=3) and 0.151 (d=5), so all grid points beat break-even in this model.
  – Located the gate-error limit: d=3 ratio 0.985 at p_gate=0.03 and 2.16 at 0.06; d=5 reached 1.90 only at 0.1; history-free decoding gave 0.393 vs 0.134 (d=3).
- **Measurement-reduced erasure recovery on QEC codes** *GitHub: erasure-local-stabiliser-measurements | Oct 2026*
  Stack: Python, NumPy, pandas, Matplotlib, pytest, GitHub Actions.
  – Built a bit-packed GF(2) erasure decoder for toric and bivariate bicycle codes (n=32-144) comparing full, local and minimal syndrome-measurement policies over 4 seeds.
  – At p=0.1, local measurement needed 67.2 of 144 stabilisers on BB-144 (47%) and 43.9 of 128 on toric-128 (34%), with logical failure identical to full measurement shot for shot.
  – With syndrome-bit flips q=0.01 on BB-72, full measurement failed/flagged 0.487 of shots versus 0.270 for local measurement; the minimal subset gave 0.130 silent failures.
- **Rydberg-EIT RF electrometry: model fit vs VQC estimators** *GitHub: rydberg-eit-rf-estimation-vqc-vs-fit | Oct 2026*
  Stack: Python, NumPy, PyTorch, pandas, Matplotlib, pytest, GitHub Actions.
  – Built a 4-level Doppler-averaged Lindblad model of Rydberg-EIT/Autler-Townes spectra; the nominal-model ML fit gives 0.057 MHz RMSE at sigma=0.02 vs a Cramér-Rao bound of 0.059 MHz (3 seeds).
  – Showed 10% unmodelled coupling-laser drift raises the fit error to 0.525 MHz, and that MLPs (0.532 MHz) and a 37-parameter 4-qubit VQC (0.788 MHz) trained on 256 spectra do not recover it.
  – Ran a VQC ablation (depth 1/2/3, no re-uploading, 1000/100-shot readout): all variants stayed at 0.591-1.105 MHz RMSE, no better than the classical baselines; CI-tested package, one-command reproduction.
- **QAOA best-of-K sampling under gate noise** *GitHub: qaoa-best-of-k-sampling-noise | Oct 2026*
  Stack: Python, NumPy, SciPy, pandas, NetworkX, Matplotlib, pytest, GitHub Actions.
  – Built an exact density-matrix QAOA simulator with depolarising noise and closed-form best-of-K metrics; ideal p=2 QAOA reaches a K=5 best-of-K cut ratio of 0.988 versus 0.792 for uniform random sampling.
  – Swept two-qubit error eps2 from 0 to 0.4 over 5 random 3-regular graphs (n=8): p=1 K=5 ratio falls from 0.946 to 0.806, approaching the random floor of 0.792, while the mean cut at eps2=0.1 is 0.718 (p=1) vs 0.702 (p=2).
  – Compared against an equal-evaluation-budget hill climber and scaled to n=10 (K=30): QAOA p=1 0.983, random 0.874, hill climbing 0.936; packaged with tests, CI and a one-command 120 s reproduction.
- **Photon-budget-aware calibration of integrated photonic meshes** *GitHub: photonic-mesh-characterisation-photon-budget | Oct 2026*
  Stack: Python, PyTorch, NumPy, pandas, Matplotlib, pytest, GitHub Actions.
  – Built a differentiable 4-6 mode Mach-Zehnder mesh simulator with beamsplitter-angle errors and exact one- and two-photon statistics; unit-tested for unitarity, normalisation and the HOM dip.
  – Compared single-photon vs two-photon maximum-likelihood characterisation at equal shot budget: two-photon gave lower post-recompilation infidelity in 27 of 30 paired cases and in 6 of 6 cases at 300 events per setting.
  – Recompiling on the fitted model cut infidelity to a median 0.038x (two-photon) and 0.083x (single-photon) of the naive ideal compile across sigma 0.03-0.10 rad, with 3 seeds and shot noise.
- **Heralded vs unheralded noise in monitored Clifford circuits** *GitHub: heralded-vs-unheralded-noise-monitored-clifford | Oct 2026*
  Stack: Python, NumPy, pandas, Matplotlib, pytest, GitHub Actions.
  – Wrote a sign-free GF(2) mixed-stabiliser simulator with uniform Sp(4,2) Clifford gates, entropies via S_A=|A|-dim(G_A), and 6 passing tests.
  – Measured the noiseless information threshold proxy p_half = 0.111 +/- 0.009 (L=16, 4 seeds) and a L=8/16 curve crossing at p = 0.1025.
  – Compared three noise channels at q=0.005: p_half 0.076 +/- 0.016 (heralded erasure), 0.056 +/- 0.015 (unheralded), 0.076 +/- 0.010 (dephasing); gap ~1.3 sigma.

- **Chip-aware QAOA placement on modular quantum devices** *GitHub: modular-chip-aware-qaoa-placement | Oct 2026*
  Stack: Python, NumPy, NetworkX, Pauli-trajectory noise simulation, pytest, GitHub Actions.
  – Studied vertex-to-chip assignment for QAOA MaxCut on two-chip modular processors where chip-to-chip coupler gates are 1x, 5x or 20x noisier than intra-chip gates (n=8-12 qubits, 4 graphs per cell, 200 noise trajectories).
  – Compared random placement, Kernighan-Lin bisection and exact minimum bisection with shared noiselessly optimised depth-2 angles, so differences come only from noise.
  – At 20x coupler error, Kernighan-Lin raised the approximation ratio by 0.04-0.06 on average for community and 3-regular graphs (e.g. 0.794 to 0.848 for community graphs at n=8) and cut the mean coupler-gate count from 6.0 to 1.75.
  – Gain was exactly 0 at equal error (control) and 0.009-0.018 at 5x; simulation only, no hardware or quantum-advantage claim.
- **Break-even noise for constrained XY-mixer vs penalty QAOA** *GitHub: xy-mixer-vs-penalty-qaoa-noise-crossover | Oct 2026*
  Stack: Python, NumPy, SciPy (COBYLA), NetworkX, Pauli-trajectory noise simulation, pytest, GitHub Actions.
  – Measured where the extra two-qubit gates of a constrained XY-ring mixer with Dicke start cancel its benefit over penalty QAOA on densest-k-subgraph (n=8, k=4, 5 random graphs), using a CX-count-weighted depolarising noise model.
  – Depth-2 XY-Dicke QAOA fell from a feasible-weighted approximation ratio of 0.758 (noiseless) to 0.390 at two-qubit error p2=5e-3 and 0.295 at 1e-2, against 0.349 noiseless for penalty QAOA, so the crossover lies between p2=5e-3 and 1e-2.
  – XY-Dicke stayed above penalty QAOA at every tested noise level, but the gap at p2=2e-2 (0.207 vs 0.176) is within one standard deviation.
  – Ablation: XY mixer with a uniform start reached only 0.214 at p2=0, so the Dicke initial state drives the advantage; simulation only.
- **Noise-aware co-design of frequency conversion for heterogeneous quantum links** *GitHub: qfc-codesign-heterogeneous-quantum-links | Oct 2026*
  Stack: Python, PyTorch (autograd, Adam), NumPy, SciPy, pandas, pytest, GitHub Actions.
  – Built a differentiable model of a heralded two-node link (930 nm broadband emitter and 780 nm narrowband memory) with PPLN quantum frequency conversion, Raman pump noise, fibre loss and memory dephasing; the objective is the hashing-bound distillable-entanglement rate.
  – Validated the model against a Fock-space Bell-state-measurement simulation (max error 3.3e-16 in fidelity), numerical quadrature of the overlap integral, and an event-level Monte Carlo (max |z| = 2.03).
  – Co-design over 10 randomised hardware instances with 10% calibration error gave 10.9 +/- 4.6 ebit/h at 100 km versus 2.68 +/- 1.06 for the O-band standard (median gain +312%, 10/10 wins, Wilcoxon p = 0.002); the C-band standard gave zero.
  – Ablation showed wavelength choice (a target near 1440-1470 nm) accounts for almost the whole gain; simulation study with representative hardware parameters, not an experimental demonstration.

Ph.D. & core portfolio projects

- **Hardware-Efficient-QPU-Ansatz | VQE for H_2** *GitHub: Hardware-Efficient-QPU-Ansatz*
  – Implemented VQE for the hydrogen molecule with a hardware-efficient ansatz tailored to Rigetti-style QPU topology.
  – Used native gate sets (Ry, Rz) and nearest-neighbor CNOT constraints; converged to -1.114 Hartree.
- **QEM-ZeroNoise-Extrapolation | Quantum Error Mitigation** *GitHub: QEM-ZeroNoise-Extrapolation*
  – Developed ZNE protocols using circuit folding and Richardson extrapolation on a custom density-matrix simulator.
  – Reduced expectation-value error from 31.5% to 5.1% under simulated depolarizing noise models.
- **QML-TensorNet-ManyBody | MPS Simulation** *GitHub: QML-TensorNet-ManyBody*
  – Built a Matrix Product State (MPS) solver from scratch to simulate the 1D Transverse-Field Ising Model.
  – Analyzed the quantum phase transition at $h = 1.0$, matching exact diagonalization for many-body systems.
- **VQA-Quantum-Dynamics | Trotterized Simulation** *GitHub: VQA-Quantum-Dynamics*
  – Translated Heisenberg XY Hamiltonian dynamics into Trotterized gate circuits using PennyLane.
  – Simulated magnon propagation in a 4-qubit spin chain starting from a Néel state.
- **QML-Advantage-Benchmarking | QAOA vs Classical** *GitHub: QML-Advantage-Benchmarking*
  – Benchmarked QAOA against brute-force MaxCut solvers to define the "crossover regime".
  – Quantified approximation ratios and runtime scaling across graph sizes ($N = 4$ to $N = 10$).

PROFESSIONAL EXPERIENCE

Assistant Professor, Computer Science | Raghu Engineering College *Aug 2024 – Nov 2024*

- Taught AI/ML courses; mentored undergraduate projects in quantum and data science; developed new curriculum.

Backend Web Development Intern | PMN Company/Chegg *Jun 2023 – Aug 2023*

- Built Django/Flask services for education platforms, integrated REST APIs, and optimized database queries.

EDUCATION

- **Ph.D. in Computer Science (Pursuing)** | SRM University AP, India *Ongoing*
  – *Research Focus:* Quantum Machine Learning and QAOA Optimization.
- **M.Tech in Data Science (Gold Medalist)** | JNTU-Gurajada *2022 – 2024*
- **B.Tech in Mechanical Engineering** | Siddharth Institute of Engineering *2019 – 2022*

CERTIFICATIONS & HONORS

- **Gold Medalist** in M.Tech Data Science for academic excellence.
- **Quantum Algorithms & Cryptography** | NPTEL (IIT Madras).
- **Deep Learning** | NPTEL.
- **Python for Data Science** | NPTEL.

CONTACT

- **GitHub:** github.com/pradeepdegalaroyal
- **LinkedIn:** linkedin.com/in/degala-chenchupradeep-811916176
- **Website:** pradeepdegalaroyal.github.io
- **Email:** Pradeepdegala143@gmail.com